SCHOOL OF COMPUTER STUDIES
Department of Computer Science and Information Technology

COURSE NUMBER	**PROG2**
COURSE TITLE	**COMPUTER PROGRAMMING 2**
COURSE UNITS	**3.0**
PRE-REQUISITES	**COMPUTER PROGRAMMING 1 (PROG 1)**
TERM OFFERED	**FIRST YEAR - SECOND SEMESTER**

Catholic Attributes

1. Orthodoxy in Catholic Doctrine 2. Integrity in Moral Life 3. Dynamism in Prayer Life

Augustinian Recollect Attributes

1. Contemplative 2. Communitarian 3. Apostolic 4. Marian

USJ-R's Vision

We envision the University of San Jose-Recoletos to be a premier Gospel and Community-oriented educational institution committed to lead in instruction, research, community engagement, and innovation in order to transform Josenians into proactive and compassionate leaders, creators of communion, and dynamic partners of society in the 21st century.

Mission

We are a Catholic University imbued with the spirit of "Caritas et Scientia", committed to providing the highest level of Quality Christian Community-oriented Education and instilling a culture of continuous learning, life-sharing, multi-disciplinary orientation, pioneerism, discovery, and innovation.

Core Values

1. Interiority 2. Nationalism 3. Service 4. Pioneerism 5. Integrity 6. Reliability 7. Excellence

Quality Statement

USJ-R stands for quality Christian community-oriented education. In pursuit of this commitment, we continually enliven the curriculum with Gospel values, hybridize programs for value innovation, engage in interdisciplinary approach to integral formation, exemplify the Augustinian Recollect charism and core values, and leverage resources for optimal results.

Josenian Graduate Attributes

1. *Socially Responsible Communitarian 2. God-Centered Individual 3. Highly Competent Professional 4. Effective Communicator 5. Adaptive Life-Long Learner*

COURSE DESCRIPTION

This is an advanced course in C programming. This is for students who have a solid understanding of the basics of C programming. This course prepares the students to master the complexity of writing their own functions. The topics include multi-dimensional arrays, pointers, strings, structs and abstraction. Students will also learn how to optimize their code for performance, and apply principles of good software design.

PROGRAM INTENDED LEARNING OUTCOMES

Knowledge for Solving Computing Problems (IT02): An ability to apply knowledge of computing, mathematics (including statistics) and social sciences to innovate techniques in critically solving the computing needs of business, government, healthcare, schools, and other kinds of organization.

Problem Analysis (IT04): Identify, analyze and formulate innovative and efficient solution to a computing problem according to the standards of the IT profession.

Design/Development of Solutions Design (IT06): implement and deploy infrastructure solutions anchored on the best practices and standards of the IT industry scalable enough to cope with the continuous demands of the society.

COURSE INTENDED LEARNING OUTCOMES (CILO)	CILO – PILO Mapping
1. Simulate the behavior and detect the output of program codes involving the fundamental programming constructs.	IT04
2. Choose appropriate conditional and iteration constructs for a given programming task.	IT06
3. Apply the techniques of structured (functional) decomposition to break a program into smaller pieces.	IT06
4. Describe the mechanics of parameter passing and the issues associated with scoping.	IT06
5. Design, implement, test, and debug a program that uses each of the following fundamental programming constructs: basic computation, simple I/O, standard conditional and iterative structures, and the definition of functions.	IT02, IT04, IT06

COURSE OUTCOMES PER TERM

PRELIM	An application implementing user-defined functions
MIDTERM	An application implementing single-dimensional and multi-dimensional arrays
FINAL	A library of functions An application implementing data abstraction

Time Frame	Learning Competencies	TOPIC – CILO Mapping 3-Substantially Addressed; 2-Moderately Addressed 1-Slightly Addressed			Assessment (Learning Competencies)	Learning Activities	Content	Resources
		1	2	3				
PRELIM								
1 Week	• Develop programs for solving computing problems that require selection and repetition structures • Evaluate code segments and determine their output	C1		C5 C2 C1	Programming Exercises using selection and repetition statements	Program Development Code Simulation	A. Programming Control Structures Selection Repetition	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Selection and Repetition structures • List of programming problems • Dev C
3 Weeks	• Differentiate and apply different methods of parameter passing appropriate to the problem at hand		C1		Programming Exercises using functions	Program Development Code Simulation	B. Functions Void Functions Value Returning Functions	• LCD Panel/Projectors • PPT Slide Presentation

	Develop programs that implement modularization to facilitate code maintainability and minimize code redundancy			C1			Parameter Passing Recursion	• Notes on Functions and Recursion • List of programming problems • Dev C
1 week	• Write an array declaration appropriately based on a given problem • Evaluate code segments and determine their output • Develop programs that involve array manipulation	C5 C5		C1	Written test Programming Exercises using single-dimensional array	Problem Evaluation Code Simulation Program Development	C. Single-Dimensional Arrays Array manipulation	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Single-Dimensional Arrays • List of programming problems • Dev C
MIDTERM								
2 weeks	• Develop programs involving two-dimensional arrays as parameters to functions		C3	C5	Programming Exercises using multi-dimensional array	Program Development Code Simulation	D. Multi-Dimensional Arrays Array processing Arrays as function parameters	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Multi-Dimensional Arrays

								• List of programming problems • Dev C
1 week	• Determine the output of the expressions that make use of pointers • Design and develop programs for solving problems that make use of pointers		C1	C5	Programming Exercises using pointers	Program Development Code Simulation	E. Pointers Pointer variables Expressions with pointers	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Pointers • List of programming problems • Dev C
FINAL								
3 weeks	• Evaluate code segments containing strings and determine their output • Develop programs that manipulate strings passed as parameters to functions			C1 C5	Programming Exercises on string manipulation	Program Development Code Simulation	F. Strings Pre-defined string functions String manipulation	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Strings • List of programming problems • Dev C
2 weeks	• Identify and declare appropriately the elements of a given entity	C5			Programming Exercises on creating an array of structs	Program Development	G. Structs	• LCD Panel/Projectors

	• Develop programs that create an array of structs and manipulate their data			C5			Struct definition Array of structs	• PPT Slide Presentation • Notes on Structs • List of programming problems • Dev C
2 weeks	• Create a library of functions • Implement data encapsulation and abstraction		C4	C5	Programming Exercise on creating a library of functions Programming exercises on data abstraction	Program Development	H. Abstraction Procedural Abstraction Data Abstraction	• LCD Panel/Projectors • PPT Slide Presentation • Notes on Abstraction • List of programming problems • Dev C

GRADING SYSTEM

Grading Standard: Progressive Averaging Method

CLUSTER 2

Class Standing ____ **70%**

Content Knowledge ____________________________ 40%

Assignments, Workshop, seatwork (30%)

Quizzes, Oral Recitation (70%)

Performance Tasks ____________________________ 60%
- Programming Exercises (20%)
- Practical Exam (50%)
- Project (30%)

Term Exam ____ **30%**

REFERENCES

Books

3G E-Learning LLC. Basic computer coding : C++. New York, NY : 3G E-Learning, c2019

Das, Sumitabha. Computer fundamentals & C programming. Chennai : McGraw-Hill Education (India) Private Limited, c2018

Chua, Marissa. Fundamentals of programming : C-language. Malabon City [Philippines] : Jimczyville Publications, c2018

Dale, Nell, Chip Weems, and Tim Richards. C++ plus data structures. Burlington, MA : Jones & Bartlett Learning, c2018

Malik, D. S. C++ programming : program design including data structures. Boston, MA : Cengage Learning, c2018

Related Links:

https://www.geeksforgeeks.org/c-programming-language

https://www.javatpoint.com/c-programming-language-tutorial

https://www.tutorialspoint.com/cprogramming/index.htm

https://www.freecodecamp.org/news/what-is-the-c-programming-language-beginner-tutorial

SYLLABUS REVISION HISTORY		
i.	Syllabus Adaptation	A.Y. 2018-2019
ii.	Version Number (optional)	
iii.	Revision Dates	February 4, 2023
iv.	Prepared by	Gene P. Abello
v.	Reviewed by	Mr. Roderick A. Bandalan
vi.	Approved by	Dr. Jovelyn C. Cuizon